

Qostanai Local Proctor

Local exam signals for human review

The problem: reviewers need context for phones, absence and sustained off-screen attention.

Astana IT University

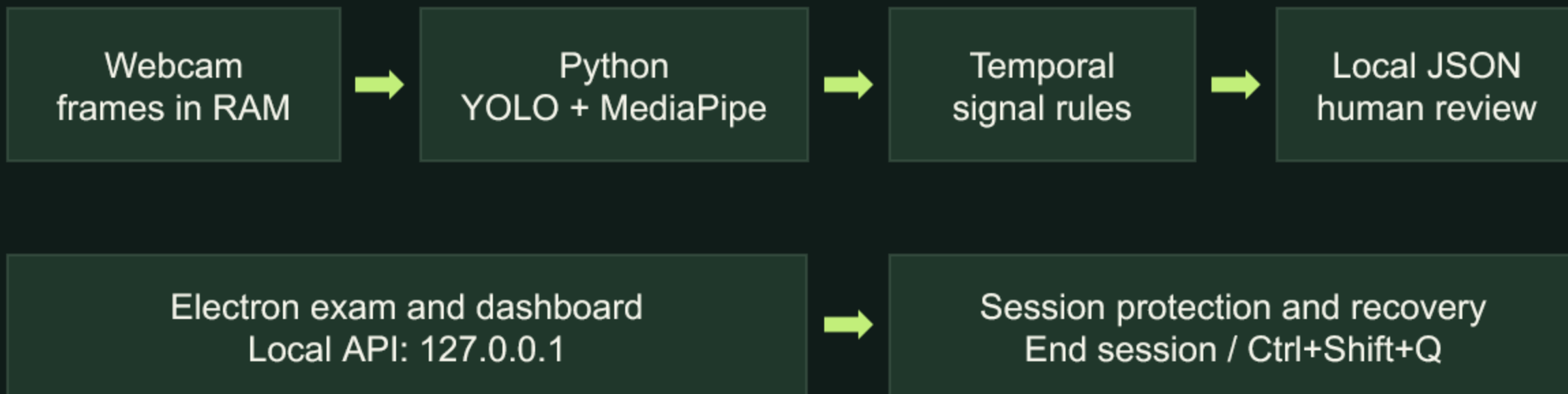
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Case requirements and coverage

Case requirement	Implemented approach	Remaining gap
Phone visible or raised	YOLOv8n + box-position rule	Photography unproven
Presence and attention	Face count + calibrated head / iris proxy	Human validation pending
Protected exam	Electron controls + optional Windows key hook	Outside-app lockdown
Functions together locally	Desktop monitor + JSON event report	Live end-to-end rehearsal

Sustained signals trigger review. Alerts do not determine misconduct.

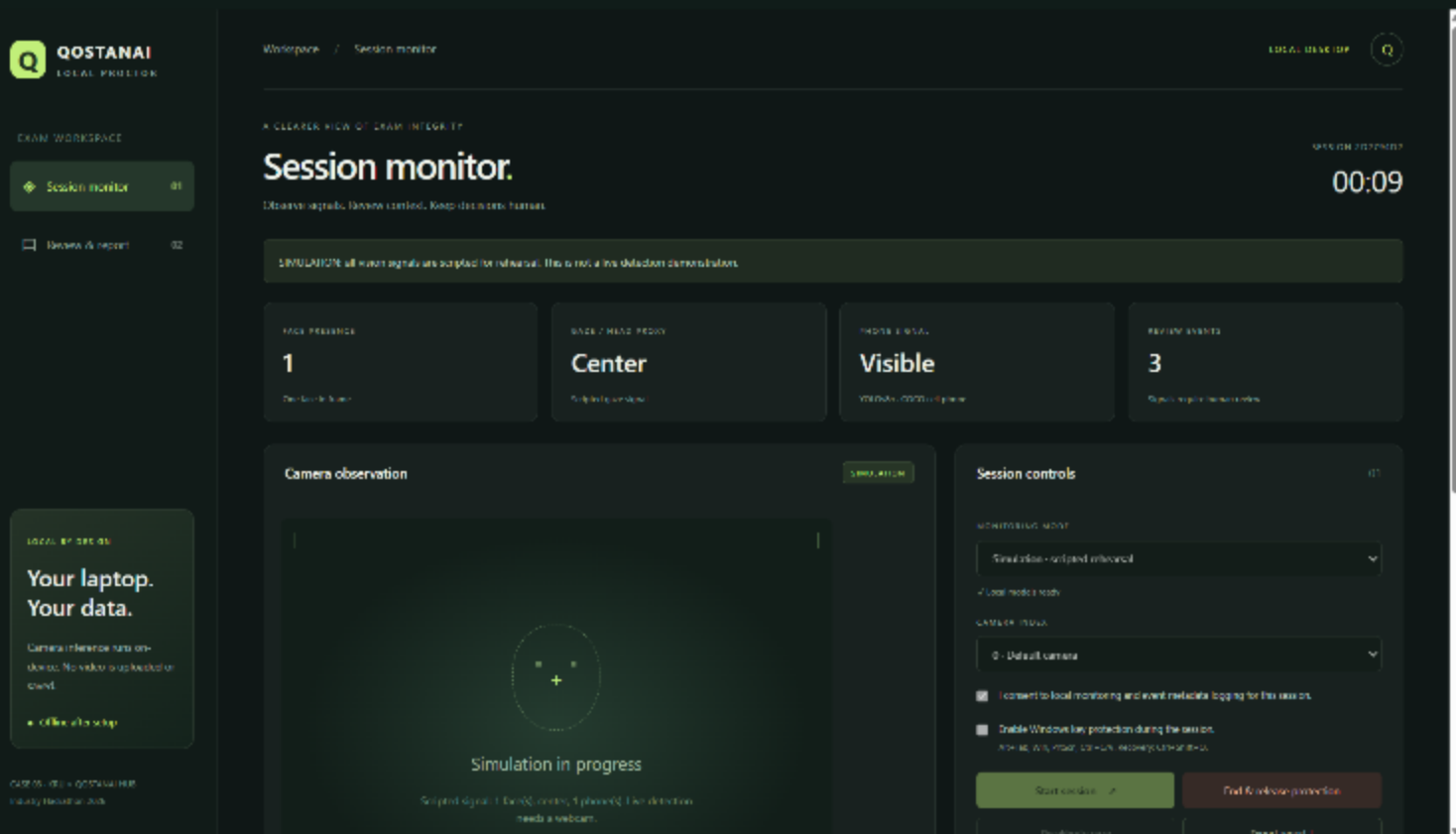
Processing on the exam laptop



Preview frames stay in memory. Reports store event metadata and, during guided tests, numeric observations and notes.

Our contribution: calibrated rules, local integration and recoverable sessions.

Prototype demonstration and evidence



26 Python checks
2 desktop policy
checks

Passed API and Electron tests

JSON download and restart
Emergency and crash release
Camera faults with fake
hardware

Human webcam trials: 0
Accuracy and live latency:
unmeasured

SIMULATION / 8 OCT 2026

Proposed supervised KRU pilot

5 volunteers 2 Windows laptops

20-minute practice sessions with a teacher present

Measure

Scenario misses and false alerts
Processing and cue-to-alert delay
Release success and reviewer time

Conditions

Consent and accommodations
Human review without penalties
Agreed storage and deletion

Hypotheses: less footage transfer and faster review. No pilot has been completed.

Limitations and next steps

Known limits

Gaze is a coarse calibrated proxy.
A raised phone cannot prove photography.
Complete OS lockdown is unfinished.
Human detection tests remain pending.

External components and AI assistance

YOLOv8n / COCO, MediaPipe, OpenCV / PyTorch, FastAPI and Electron
Codex / ChatGPT assisted development and presentation preparation.

Next: human validation, a supervised pilot and managed exam policies.